

TERM PAPER PRESENTATION
CHE 494
Graphene/2D Nanomaterials

Graphene And Graphene Based Nanomaterials For Cancer Treatment

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Presentation Overview

1

Introduction about Cancer and its Treatment

2

Challenges in current treatment methods available

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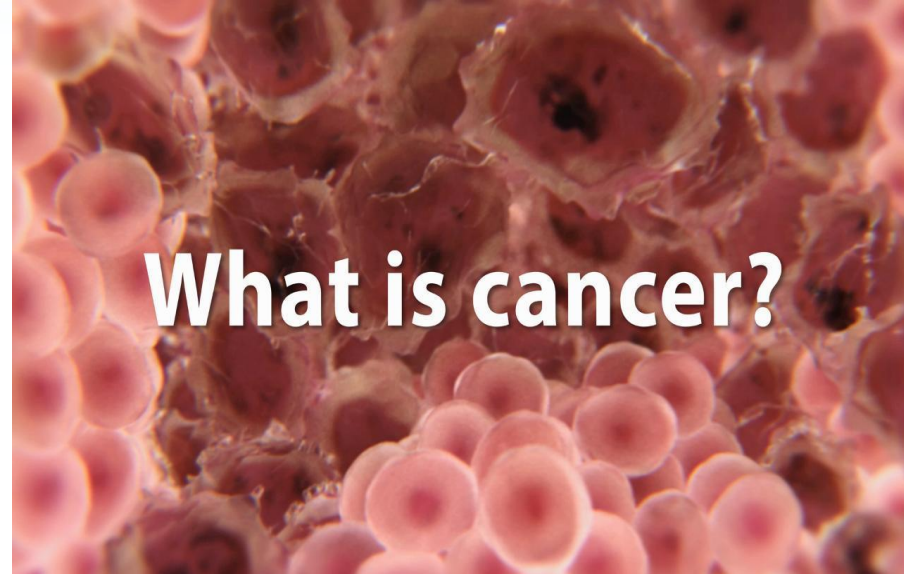
Graphene's role in Cancer Treatment

4

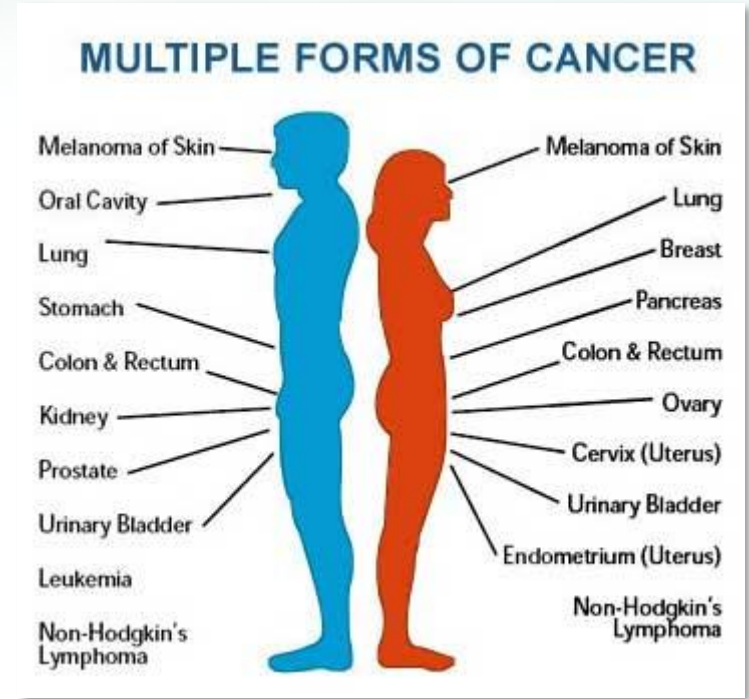
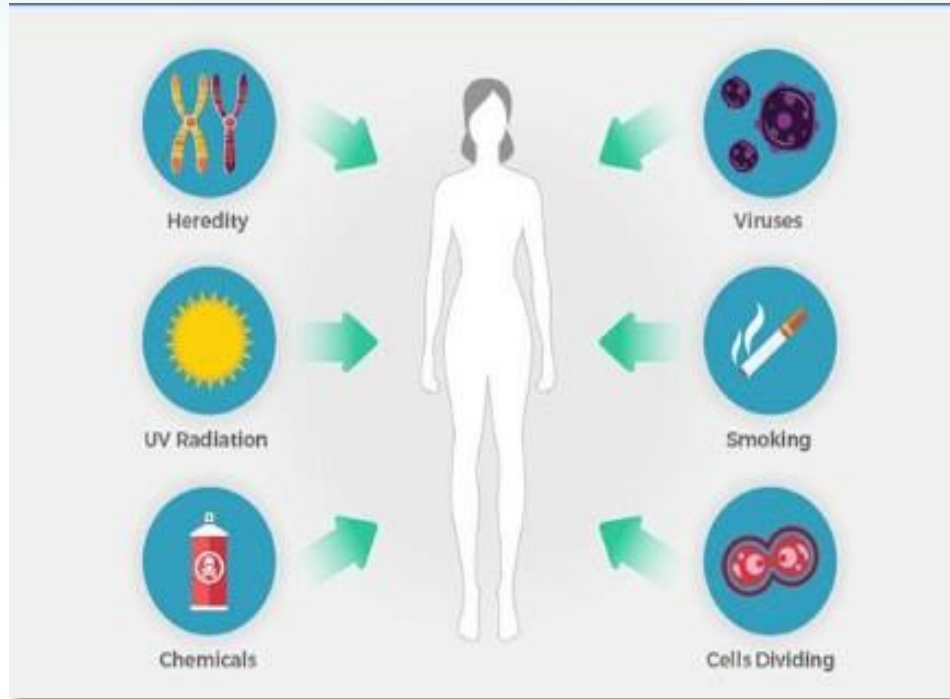
Challenges involved and Conclusions

Introduction

- Uncontrolled growth of abnormal cells in body
- Can start anywhere in the body
- Cancer- Name given to a collection of related diseases
- Extra cells spread and form tumors



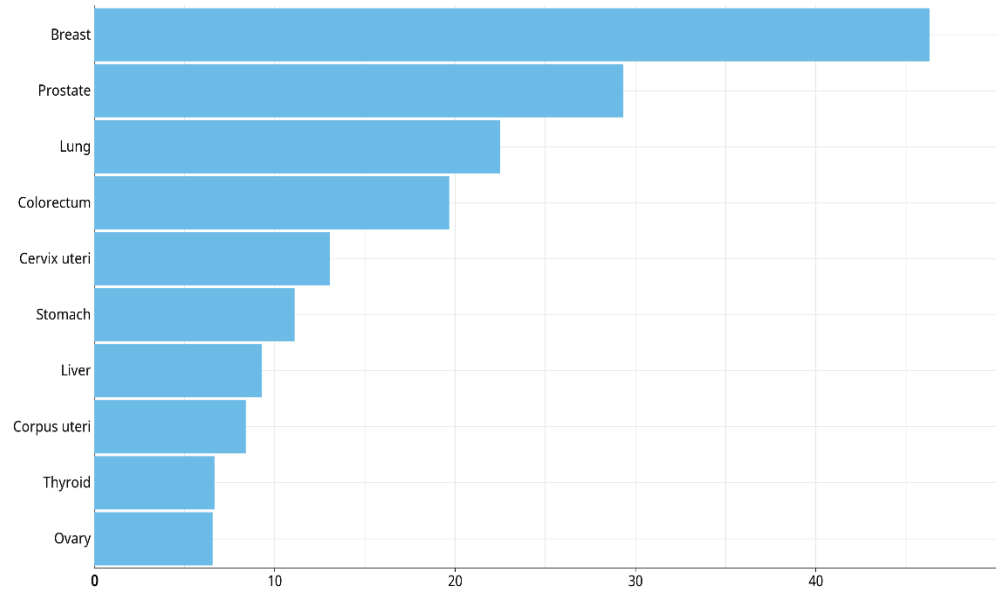
Causes and Types of Cancer



Cancer Statistics

- According to estimates from the International Agency for Research on Cancer (IARC), in 2018 there were 17.0 million new cancer cases and 9.5 million cancer deaths worldwide.

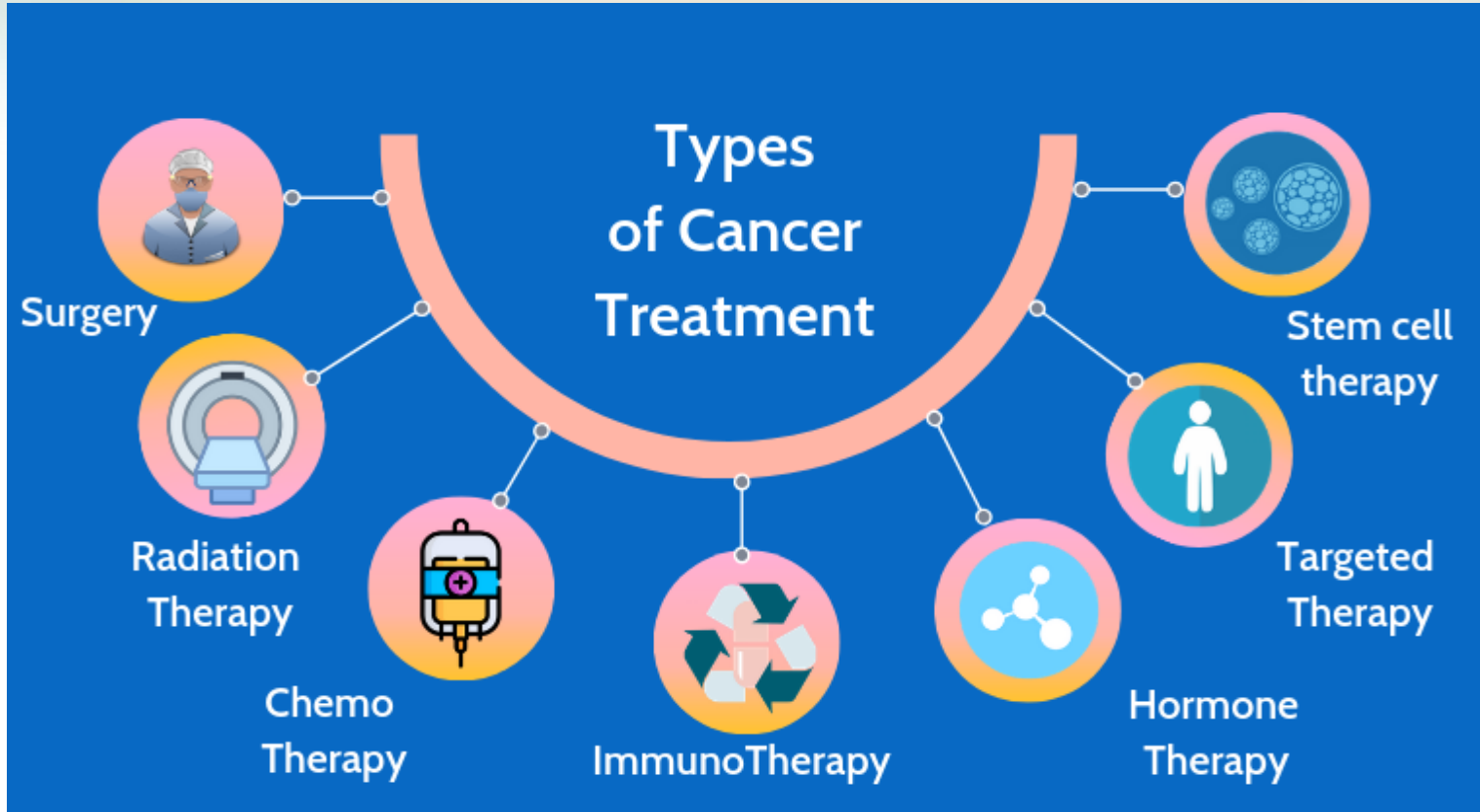
Estimated age-standardized incidence rates (World) in 2018, worldwide, both sexes, all ages



Data source: Globocan 2018
Graph production: Global Cancer
Observatory (<http://gco.iarc.fr>)

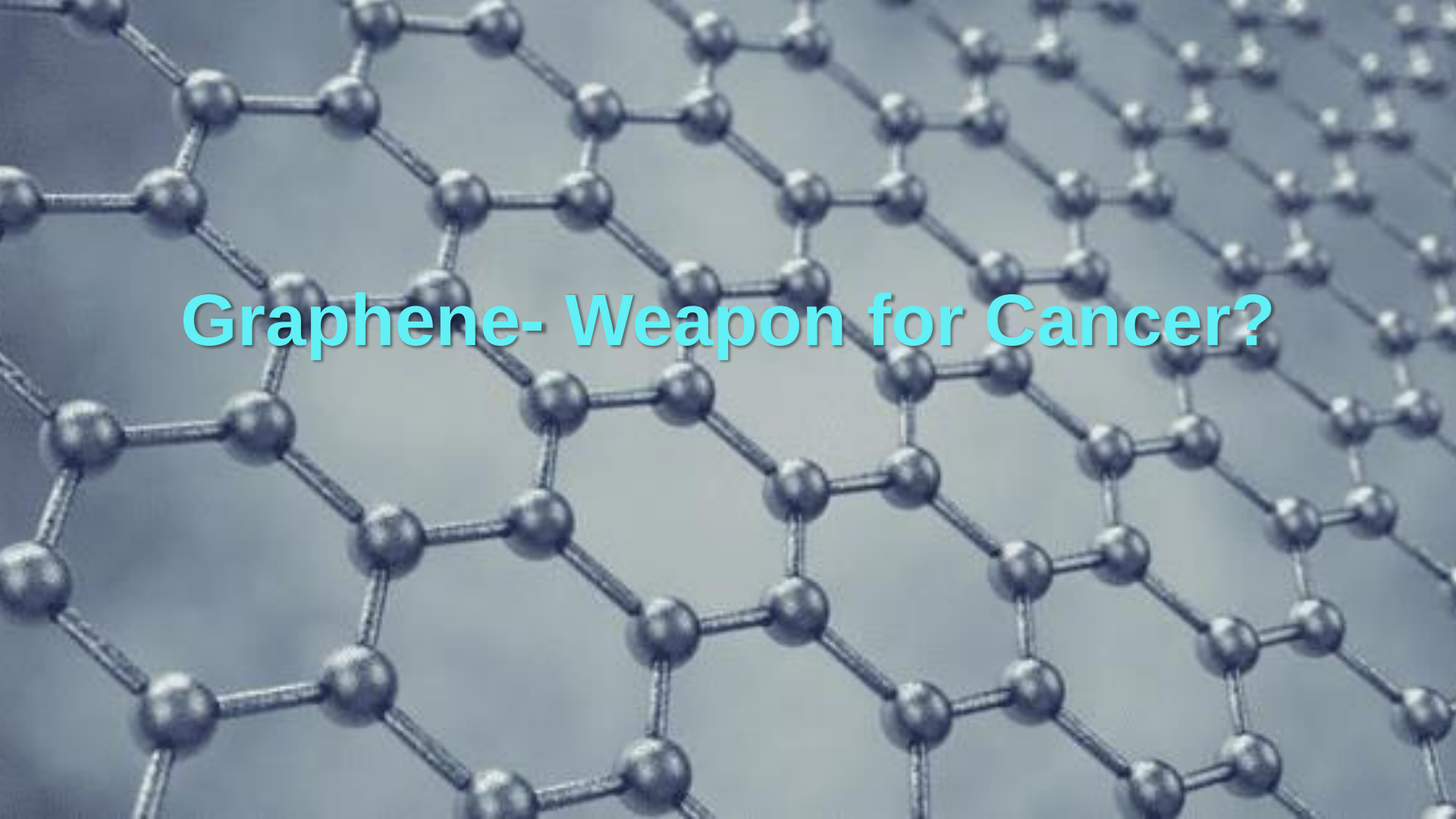
ASR (World) per 100 000

Treatment methods



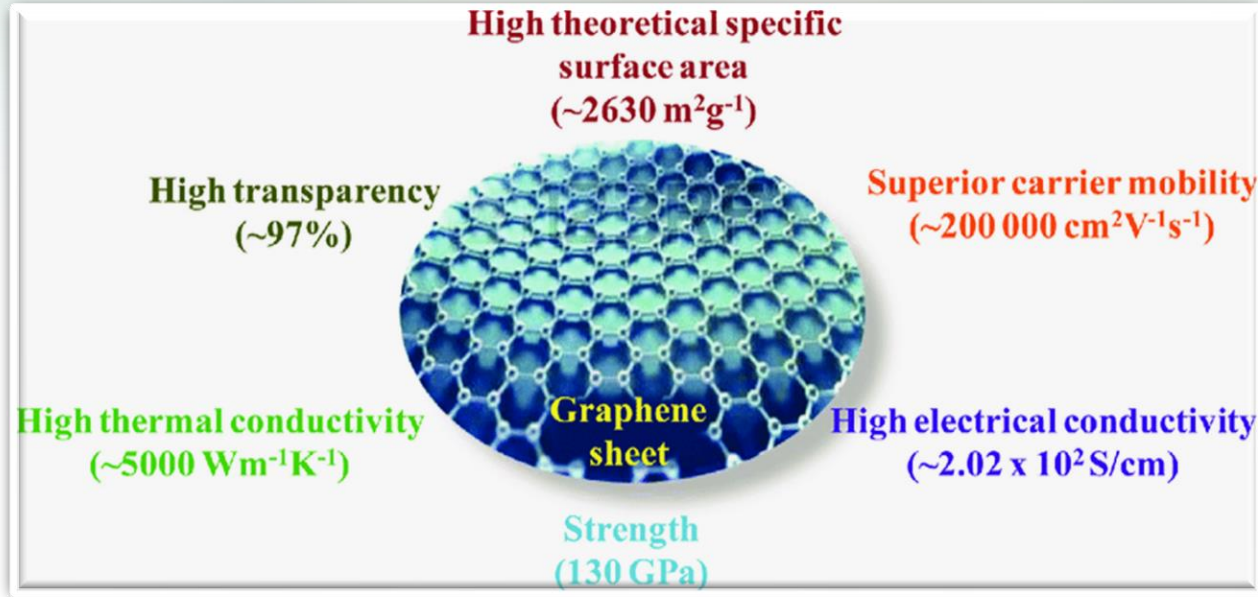
Drawbacks of current available methods

- **Immunotherapy**- Affects healthy cells and organs hence can lead to symptoms like fever, aches, pain, etc.
- **Chemotherapy**- Destroys healthy cells along with cancer cells
- **Radiation Therapy**- Major side effect: Fatigue
- **Targeted Therapy** - Cancer cells can become resistant
 - Drugs for some targets are hard to develop
- **Hormone therapy**- Can affect some body mechanisms involving hormones

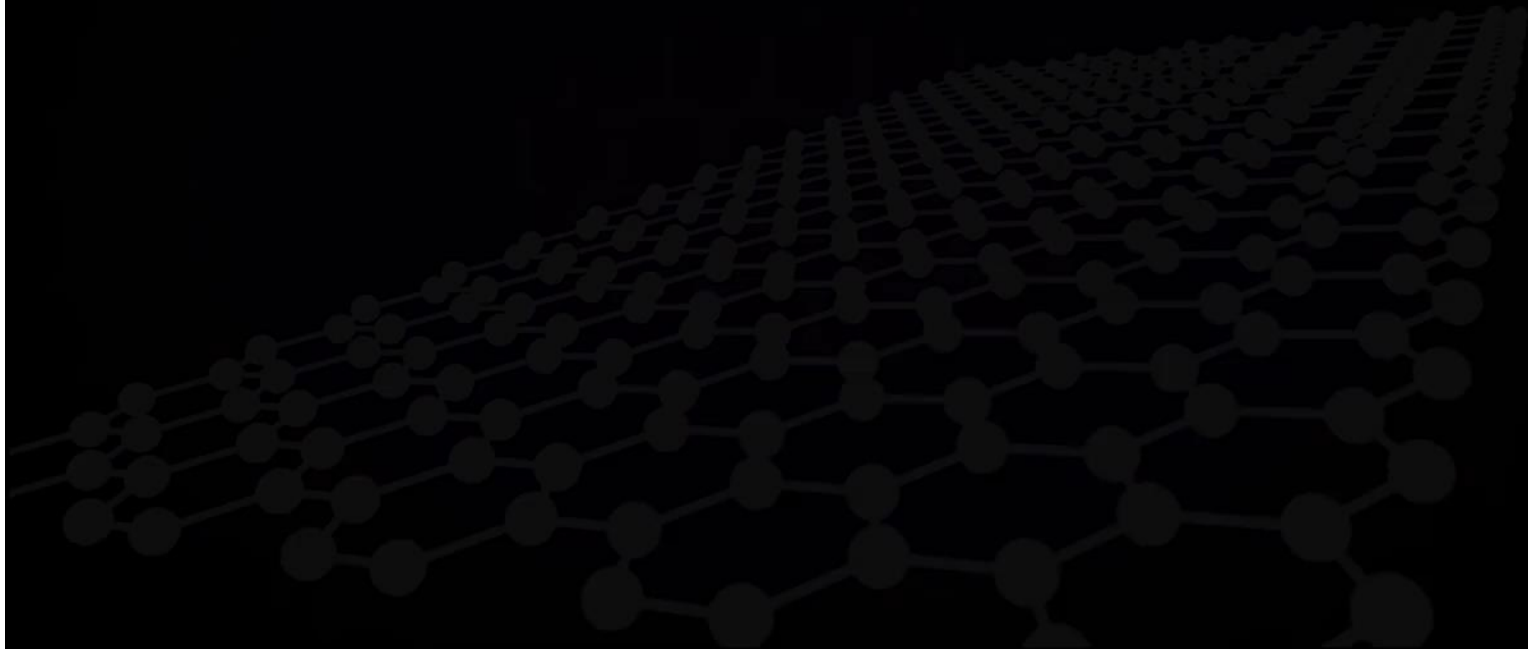


Graphene- Weapon for Cancer?

Why Graphene?



Credits-Recent Progress of Metal-Graphene Nanostructures in Photocatalysis - Scientific Figure on ResearchGate.



- Graphene is used as a platform for loading anti-cancer drugs
- Hydrophobic chemical structure and high surface area to mass ratio allows covalent and non-covalent tethering to facilitate cancer cell targeting
- Earliest report published by Liu et.al used GO covalently functionalized with PEG to load anti-cancer drugs and improved water dispersability
- This improves drug efficacy by avoiding use of organic solvent with GO which can be toxic to human body
- Zhang et al. used various combination of drugs on GO platform which showed improved efficacy.
- Recently, Folic Acid-GO, Hyaluronic acid-GO have been studied extensively for improved drug efficiency.

1. PEGylated nanographene oxide for delivery of water-insoluble cancer drugs.

Liu Z, Robinson JT, Sun X, Dai H. J Am Chem Soc. 2008 Aug 20;130(33):10876-7. doi: 10.1021/ja803688x. Epub 2008 Jul 29

2. Functional graphene oxide as a nanocarrier for controlled loading and targeted delivery of mixed anticancer drugs.

Zhang L1, Xia J, Zhao Q, Liu L, Zhang Z. Small. 2010 Feb 22;6(4):537-44. doi: 10.1002/sml.200901680

Conclusions

- Graphene shows immense potential to be used as a platform for anti-cancer drug delivery
- Novel design techniques are needed for effective encapsulation and delivery of drugs
- Development of anti-cancer drugs to be effective with GO as a carrier needs to be studied
- Still a long way from studies on human bodies because of cytotoxicity of derivatives of graphene
- Synthesis techniques, chemical composition, shape, size of Graphene also play an important role
- Various allotropes of graphene like carbon nanotubes, quantum dots should also be studied for drug delivery efficiency

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Thank you!